

SIH 1505 — INDUSTRIAL EMERGENCY PRE-PLAN

FACILITY: PetroChem Complex Alpha - Unit 04
PROTOTYPE FACILITY REF: PCH-ALPHA-04 (Demo Facility — Non-Statutory Evaluation)

INCIDENT ID: INC-T-01-14K
DOCUMENT VERSION: v0.1
[PENDING_HUMAN_AUTHORIZATION]
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PREPARED BY: SIH-1505 Decision Support Engine

INCIDENT SEVERITY: MODERATE (SCORE: 38.6/100) • GOVERNANCE: PENDING HUMAN AUTHORIZATION

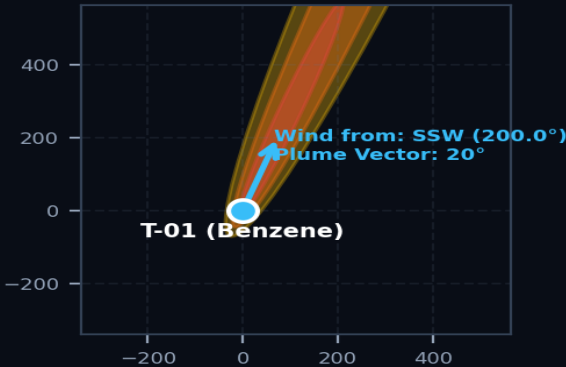
1. INCIDENT SOURCE, CHEMICAL SDS & METEOROLOGICAL PROFILE

Incident Type:	Tank Leak	Source Asset:	T-01 (21.6865, 72.5710)
Hazardous Substance:	Benzene (Pure Grade)	Release Emission Rate:	20.0 kg/s (Duration: 30 min)
Meteorological Wind Vector:	14.5 km/h from SSW (200.0°)	Plume Propagation Vector:	Downwind travel towards 20°
Ambient Temperature:	29.0°C	Weather Telemetry Source:	DEMO (Scenario Preset) • Stability: Class D

2. HAZARD THREAT ZONES (SCREENING GAUSSIAN DISPERSION ENVELOPE)

Zone Tier	Threshold Criterion	Downwind Reach	Crosswind Width	Threat Area
Red Threat Zone	ERPG-3 / High Lethality (1000.0 ppm)	664.9 m	47.4 m	20,621 m²
Orange Threat Zone	ERPG-2 / Irreversible Injury (150.0 ppm)	1930.4 m	123.6 m	155,779 m²
Yellow Threat Zone	ERPG-1 / Mild Irritation (50.0 ppm)	3000.0 m	215.5 m	445,394 m²

ATMOSPHERIC DISPERSION & THREAT ENVELOPE (Screening Gaussian Plume • T+120s)



3. POPULATION & INFRASTRUCTURE EXPOSURE IMPACT

Total Workers on Site:	28 Personnel	Lethal Zone (Red) Exposed:	0 Workers
Severe Zone (Orange) Exposed:	0 Workers	Caution Zone (Yellow) Exposed:	0 Workers
Compromised Plant Assets:	1 of 16 Units	Severed Road Segments:	4 of 10 Segments

4. DYNAMIC SAFE EVACUATION CORRIDOR & MUSTER DIRECTIVES

Designated Safe Assembly Point:	Assembly Point 3 - West Perimeter Zone	Exit Perimeter Gate:	Gate 2 - South Commercial & Tanker Logistics Gate
Total Safe Egress Distance:	373.4 meters	Estimated Walk Time:	5.2 minutes (at 1.2 m/s)
Multi-Factor Scoring:	Safety: 70% Dist: 76% Rank: 0.727	Route Clearance Status:	DIVERTED
Selection Rationale:	Selected as optimal Upwind/Crosswind egress corridor (149° clearance from plume axis) providing shortest uncompromised route to Assembly Point 3 - West Perimeter Zone (Safety Score: 0.70, Composite Score: 0.73).	Avoided Blocked Roads:	Perimeter Ring Road - West, Perimeter Ring Road - South, Chemical Sector Cross Road (East-West North), Process Area Cross Road (East-West South)

Evaluated Muster Point Alternatives & Rejection Analysis:

Candidate Assembly Point	Exit Gate	Distance	Safety	Status / Rejection Reason
ROUTE-AP-3 (AP-3)	GATE-2	373.4 m	70%	SELECTED: None
ROUTE-AP-2 (AP-2)	GATE-2	1047.1 m	70%	VIABLE_BACKUP: Internal access roads severed; requires outer peripheral detour
ROUTE-AP-1 (AP-1)	GATE-1	1569.5 m	70%	REJECTED: Internal access roads severed; requires outer peripheral detour
ROUTE-AP-4 (AP-4)	GATE-3	1784.1 m	70%	REJECTED: Internal access roads severed; requires outer peripheral detour



5. TACTICAL EMERGENCY RESOURCE DEPLOYMENT & SUPPRESSION STRATEGY

DECISION-SUPPORT RECOMMENDATION — REQUIRES SITE/HSE VALIDATION: All tactical quantities, PPE ensembles, suppression demands, and response actions are prototype computational recommendations and must be validated against facility ERDMP and competent safety authorities before operational deployment.

Upwind Staging Standoff:	250.0 meters	Isolation Cordon Radius:	831.1 meters
Firewater Demand:	6,700 LPM	Foam Concentrate Demand:	2,500 Liters (AFFF 3%)
Mandatory Entry PPE:	Level A Fully Encapsulated Gas-Tight Suit (Trellechem) with positive-pressure SCBA (NIOSH Certified)	Calculation Basis:	Calculated from release rate 20.0 kg/s, 1 vulnerable assets, and Benzene (Pure Grade) SDS toxicity/flammability

Unit Name & Type	Station	Staging Geolocation	Transit	ETA	Priority
Fire Tender Alpha (10,000L Multi-Purpose Water/Foam) Vapor Cloud Inerting & Preemptive Foam Blanket	FS-01	Upwind Incident Staging Post (Bearing 20°, 250m Standoff)	736.2 m	2.5m	IMMEDIATE
Hazmat Rapid Response Unit (Level A/B Ensembles) Hot Zone Isolation & Emergency Valve Clamping	FS-01	Incident Hot-Zone Control Post (20° Upwind)	726.4 m	2.5m	IMMEDIATE
Industrial Emergency Response Team (ERT Alpha) Cordon Enforcement, Road Junction Barricading & Headcount Control	CR-01	Main Outer Access Gate & Security Command	303.6 m	1.3m	SUPPORT

High-Volume Water Curtain Bowser Secondary Water Supply & Runoff Dilution	FS-01	Crosswind Knockdown Sector (110° Vector)	463.8 m	1.8m	SUPPORT
Advanced Cardiac & Toxic Resuscitation Ambulance Precautionary Medical Standby at Muster Point	MED-01	Medical Triage at Assembly Point 3 - West Perimeter Zone	805.5 m	2.7m	STANDBY

6. INCIDENT COMMAND EXECUTIVE SUMMARY SNAPSHOT

Composite Risk Score:	38.6/100 (MODERATE)	Chemical Released:	Benzene (Pure Grade)
Casualty Threat Matrix:	Red: 0 Orange: 0 Yellow: 0	Facility Infrastructure:	1 Compromised Assets 4 Blocked Roads
Designated Safe Muster:	Assembly Point 3 - West Perimeter Zone via Gate 2 - South Commercial & Tanker Logistics Gate	Egress Metrics:	373.4 m (~5.2 min walk)
Lead Tactical Resource:	Fire Tender Alpha (10,000L Multi-Purpose Water/Foam) (IMMEDIATE, ETA: 2.5m)	Firewater & Suppression:	6,700 LPM Water 2,500 L AFFF Foam

7. STANDARD OPERATING PROCEDURE (SOP) PHASED ACTION DIRECTIVES

Phase 1: Initial Alarm, Isolation & Personnel Evacuation (0 - 5 Minutes) - Sound Plant Emergency Siren (Continuous Alarm for Benzene (Pure Grade) Tank Leak). - Actuate Emergency Shutdown (ESD) for T-01 and isolate upstream pipeline valves. - Enforce mandatory crosswind evacuation of all Sector personnel towards Assembly Point 3 - West Perimeter Zone. - ERT Team establishes 831.1m initial cordon perimeter around T-01. Phase 2: Source Mitigation & Tactical Exposure Protection (5 - 15 Minutes) - Position Fire Tender & Bowser at Upwind Staging Post (250m standoff, bearing 20°). - Establish 6,700 LPM water curtain coverage to suppress Benzene (Pure Grade) vapors and cool 1 adjacent assets. - Deploy Hazmat Entry Squad in Level A Fully Encapsulated Gas-Tight Suit for hot-zone valve isolation and clamp seal. - Verify automated fixed water deluge systems on neighboring storage vessels. Phase 3: Containment, Medical Triage & Incident De-escalation (15 - 30 Minutes) - Complete headcount verification at Assembly Point 3 - West Perimeter Zone (Reconcile against 0 exposed workers). - Ambulance team initiates emergency triage and decontamination for exposed operators. - Contain all contaminated firefighting water in industrial effluent retention sump; prevent municipal drain escape. - HSE Lead Commander compiles situation briefing for District Disaster Management Authority (DDMA).
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8. DATA TRACEABILITY & SCREENING MODEL ASSUMPTIONS

Dispersion Physics:	Screening Gaussian Approximation with Pasquill-Gifford stability parameters	Human Governance:	Mandatory HSE human review and authorization required prior to operational execution
Weather Telemetry:	DEMO Telemetry (Scenario Preset) • Recorded at 29.0°C, 14.5 km/h	Statutory Scope:	Decision-support prototype only; non-certified under statutory OISD/ALOHA rules
Terrain Modeling:	Simplified 2D/3D flat terrain representation with standard surface roughness	Road & Worker Graph:	Prototype facility CAD/GIS network with obstacle avoidance pathfinding

9. REGULATORY DISCLAIMER & PROTOTYPE NOTICE

PROTOTYPE DECISION SUPPORT — NON-CERTIFIED COMPUTATIONAL WORKFLOW: This emergency pre-plan document is prepared automatically by the SIH 1505 Decision Support Engine. Dispersion envelopes, routing corridors, and tactical dispatches are computed for hackathon demonstration and decision-support prototyping only. This software does NOT claim certified ALOHA equivalence or statutory compliance. Operational tactical actions require validation against the facility's approved Emergency Response and Disaster Management Plan (ERDMP), applicable OISD/PESO standards, and competent statutory safety authorities.

10. HUMAN-IN-THE-LOOP PRE-PLAN AUTHORIZATION & ENDORSEMENT

DOCUMENT STATUS: PENDING HUMAN AUTHORIZATION (Version v0.1) Prepared by: SIH-1505 Decision Support Engine Approver: Not Assigned (Pending Human Review) Role: Not Assigned Authorization ID: None (Draft State) Human Authorization: REQUIRED <i>Document is in draft screening stage until human authorization is completed.</i>	HUMAN AUTHORIZATION SIGNATURE: _____ HSE Controller / Incident Commander HUMAN AUTHORIZATION REQUIRED — NO SIGNATURE
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